Evaluating Generative AI Workflows

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Great Learning
POWER AHEAD

Let’s begin the discussion by answering a few questions.

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

Which evaluation metric is robust to paraphrases because it compares meaning using contextual, embedding-based similarity?

A. BLEU

B. ROUGE

C. Exact Match

D. BERTScore

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

Great Learning
POWER AHEAD

Which evaluation metric is robust to paraphrases because it compares meaning using contextual, embedding-based similarity?

A BLEU

B ROUGE

C Exact Match

D BERTScore

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluation Metrics - Lexical

BLEU

Measures **n-gram** overlap between candidate and reference with a brevity penalty.

Introduced and widely used for machine translation tasks

Favors exact word overlap, so it underscores good paraphrases

ROUGE

Recall-focused metric checking how much of the reference's n-grams appear in the candidate.

Standard choice for evaluating summaries.

Variants (ROUGE-1, -2, -L) capture unigram, bigram, and sequence-level overlap.

Exact Match

Binary metric: score 1 if output string matches reference exactly, else 0.

Common in tasks like extractive QA or structured prediction with fixed answers.

Too strict for open-ended generation where many valid phrasings exist.

Lexical metrics reward output that LOOKS like the reference - not output that MEANS the same thing. Useful as baselines, never primary.

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluation Metrics - Semantic

BERTScore

Contextual, embedding-based similarity between candidate and reference using BERT

Compares meaning rather than exact words

More robust to paraphrases than lexical metrics, but can be sensitive to choice of model and language coverage

COMET

COMET (Crosslingual Optimized Metric for Evaluation of Translation) is a Neural evaluation model (usually for Machine Translation)

It scores quality using source, hypothesis, and sometimes reference.

Trained on human judgments, it better aligns with human perception of adequacy and fluency than pure overlap-based metrics

LLM-as-a-Judge

Reasoning-based evaluation using a large language model to score outputs based on a natural language rubric.

Evaluates complex dimensions like logic, instruction-following, and tone without requiring a reference

Highly aligned with human reasoning and nuance, but susceptible to biases which might lead to unfair or inaccurate scores

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluation Metrics - Lexical vs Semantic

Paraphrase

Reference: "Revenue grew 12% YoY."

Output: "YoY revenue increased by twelve percent."

Lexical: LOW (few n-grams)

Semantic: HIGH (same meaning)

Added word

Reference: "The deployment succeeded."

Output: "The deployment has not succeeded."

Lexical: HIGH (n-gram overlap)

Semantic: OPPOSITE (meaning is inverted)

Exact match

Reference: "Paris is capital of France."

Output: "Paris is capital of France."

Lexical: PERFECT

Semantic: PERFECT

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

A prompt described as "stable but sensitive" indicates what two distinct diagnostics about its production readiness?

A. The model is too random, but the output is consistent across runs

B. Output is consistent across repeated runs, but small semantically equivalent tweaks cause significant change

C. The prompt is brittle, but the model + parameter combination is reliable

D. The prompt passed quality checks in a single run, but variance is unknown

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

A prompt described as "stable but sensitive" indicates what two distinct diagnostics about its production readiness?

A. The model is too random, but the output is consistent across runs

B. Output is consistent across repeated runs, but small semantically equivalent tweaks cause significant change

C. The prompt is brittle, but the model + parameter combination is reliable

D. The prompt passed quality checks in a single run, but variance is unknown

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Prompt Stability vs Prompt Sensitivity

LLM outputs vary both across repeated runs and with small prompt changes
Both must be measured.

📊 Stability

Method:

- Run SAME prompt multiple times
- Measure output consistency across runs

Analysis:

- Low stability → model unreliable at this temperature
- Diagnoses: model + parameter combination

🔬 Sensitivity

Method:

- Make small, semantically equivalent tweaks
- Measure how much output changes

Analysis:

- High sensitivity → prompt is brittle
- Diagnoses: prompt design

A prompt can be stable but sensitive.
You need BOTH to know it's production-ready.

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

A team notices that their model's performance plateaus despite adding more training data. Upon inspection, different annotators have labelled the same examples differently. What does this most likely indicate according to IAA principles?

A. The model has reached its architectural limit and needs a larger architecture

B. Low IAA means the rubric is unclear or the task is ambiguous, and the model is learning noise rather than signal

C. The gold set is too small and needs more examples added

D. The annotators need more domain expertise before labelling

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Evaluating Generative AI Workflows Quiz

A team notices that their model's performance plateaus despite adding more training data. Upon inspection, different annotators have labelled the same examples differently. What does this most likely indicate according to IAA principles?

A. The model has reached its architectural limit and needs a larger architecture

B. Low IAA means the rubric is unclear or the task is ambiguous, and the model is learning noise rather than signal

C. The gold set is too small and needs more examples added

D. The annotators need more domain expertise before labelling

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Inter-Annotator Agreement (IAA)

If two trained humans can't agree on a label, no model trained on those labels, and no evaluator scoring against them can be trusted.

Data Quality Signal

Is the gold set actually gold?

- High IAA = labels are reliable
- Low IAA = rubric unclear or task is ambiguous
- Noisy labels causes the model to learn the noise

Rubric Diagnostic

Disagreement points at the fix

- Low-IAA items expose missing edge-case rules
- Forces a real definition of each label
- Improve the rubric, not the annotators

Model Ceiling

Humans cap what a model can achieve

- A model can't reliably exceed human agreement
- Reporting accuracy without IAA is misleading

Measure IAA BEFORE you trust the dataset, NOT after you've trained on it

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

IAA for Diagnostics

Low agreement on easy cases

Rubric is underspecified. Annotators are applying different implicit criteria.

Low agreement on hard cases only

Usually expected and occurs when the data is genuinely ambiguous. Document the ambiguity and don't force consensus.

Drift over time

Early agreement high, later low. Signals training wear-off or changing interpretation.

Systematic bias

One annotator consistently scores higher/lower. Calibration needed.

Disagreement usually means the rubric or example is the problem - not the annotators.

This file is meant for personal use by venugopal.adep@gmail.com only.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Q&A

This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is strictly prohibited.
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Great Learning

Power Ahead !



This file is meant for personal use by venugopal.adep@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.